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Exp no: 1

BASIC NETWORK COMMANDS

AIM:
Study of various Network commands used in Linux and Windows

1 arp -a

ARP - Address Resolution Protocol. It will show the IP address of your computer along with the IP address and MAC address of your router

OUTPUT: - gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on eth0

2 hostname

This is the simplest of all TCP/IP commands It simply displays the name of your computer

OUTPUT: localhost: local domain.

3. ipconfig/all:

This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of Ethernet adapter in your system.

4. nbt stat -a

This command helps solve problems with Net BIOS name resolution. (Nbt → NetBIOS over TCP/IP)

5 netstat [Network statistics]

Netstat displays a variety of statistics about computers active TCP/IP connections It is a command line tool for monitoring network connections both incoming and outgoing as well as routing tables interface statistics etc

OUTPUT:

Proto	Recv-Q	Send Q	Local Address	Foreign address	State
Udp	0	0	local host. localdl	gateway bootst	Estab

Proto	RefCnt	Flags	Type	State	I-Node	Path
Unix	2	[ ]	DGRAM		28018	(run/user/
d/notify						system
Unix	3	[ ]	DGRAM	connected	13684	(run/system/
						registry)

6. nslookup:

[name server lookup] is a tool used to perform DNS lookups in linux. It is used to display DNS details, such as the IP address of the particular computer, the MX records for a domain or the NS servers of a domain. nslookup can operate in 2 modes: interactive & non-interactive.

OUTPUT: Server: 172.16.8.1
Address: 172.16.8.1.53

7. Pathping:

Pathping is unique to window's and is basically a combination of the ping and tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

OUTPUT:

8. ping:

[Packet Internet Grouper] command is the best way to test connectivity between two nodes. Ping uses ICMP [Internet control Message Protocol] to communicate to other devices.

OUTPUT:

PING localhost (localhost (::1)) 56 data bytes

64 bytes from localhost (::1): icmp_seq = 1 ttl = 64 time = 0.045 ms

64 bytes from localhost (::1): icmp_seq = 2 ttl = 64 time = 0.096 ms

9. Route:

Route command is used to show / manipulate the IP routing table. It is primarily used to setup static routes to specific host or network via an interface

OUTPUT:

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use Iface
default	- gateway	0.0.0.0	UG	100	0	0enpos 3
10.0.2.0	0.0.0.0	255.255.255.0	0	100	0	0enpos3

LINUX COMMANDS.

1. ip

The ip command is one of the basic commands every administration will need in daily work, from setting up new systems and assigning IP's to troubleshooting existing systems.

The ip command can show address information, manipulate routing, plus display network various devices interfaces and tunnels.

OUTPUT:

Usage: ip [OPTIONS] OBJECT {COMMAND | help}

ip [force] - batch filename

where OBJECT := {address | route | help | ila | ioam | l2tp | link | macsec | maddress | monitor}

OPTIONS := {-V [ersion] | -s [tatistics] | -d [etails] | [-r [esolve] -h [uman-readable]}

2. ifconfig

The ifconfig command was/is a staple in many sysadmin tool belt for configuring and troubleshooting network. It has since been replaced by the ip command discussed above

OUTPUT:

enp0s3: flags = 4163 < UP, BROADCAST, RUNNING, MULTICAST >
mtc 1500 inet 10.0.2.15 netmask 255.255.255.0
broadcast 10.0.2.255 inet 6 fd 17:625c:f037:2.
a00:27ff:fef9:2eab prefix len 64 scope id
0x0 < global >

3. mtr

(Matt's traceroute) is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool. This command combines the functionality of the ping and traceroute comments. Just like a trace route, the mtr command will show the route from a computer to a specified host. mtr provides a lot of statistics about each hop, such as response time or packet loss, then obviously, there is bad link somewhere mtr google.com

OUTPUT:

localhost. localdomain (10.2.15) -> google.com
2026-07-23 to 93A

4. tcpdump:

tcpdump is command is designed for capturing and displaying packets.

OUTPUT:

5. Ping:

Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request messages. The receipt of corresponding echo Reply messages is displayed, along with round-trip times. Ping is the primary TCP/IP command used to troubleshoot connectivity, reachability and name resolution

ping google.com